PACLAND · Physics, in plain words

What we checked, what we found, and what to do next. One page, no jargon.

1 · The question

String theory allows about 10^500 different universes, each with different particle masses. How would you ever know which one is ours? The usual answer is: you can't.

2 · What we found

- There is exactly ONE exact number, and it is a total. Picture a household budget: the total spend is fixed, the individual receipts are free. The total flux is exact; how it is split between hidden dimensions is not.
- That is provable, not a hunch. Every quantity that survives the theory's symmetries is a power of that one number. We machine-checked the proof step by step, so it is “the computer verified it”, not “we believe”.
- The 19 free numbers of particle physics stay free. Nobody derives the electron mass; a final theory supplies relations between numbers, not the numbers.

3 · The best we have right now

A tower of six results that are machine-checked rather than argued, plus one workable model universe — a six-dimensional torus — where every rule about quantum gravity can be proved exactly. Counted honestly, that is nine things we can check against the real world. Not the answer to string theory; the only part of it that is not opinion.

4 · How we refine it

- Finish the proofs: most are checked only to low orders, and the general case is the next slice of work.
- Use free checks: a shape and its mirror carry swapped labels, so every mirror pair is a known answer to test the tools against — no new data, no new experiment.
- Count the deficit upward only where it can grow: the flux side is saturated at one, so the next rungs must come from branes and the Standard Model.

5 · What this means for the world

It tells physicists where a final theory can deliver numbers and where it cannot. That is worth money: a detector built to measure a number the theory can never give is a detector built on a misreading. And the method — machine-check the invariants, count what is verified, state the rest as free — is what we now apply to climate, medicines and economics.

6 · What this cannot do

It cannot say which universe is ours, or what the masses are. Nobody can, yet. What it can do is say exactly where the free choices are — and that is the part that can be checked.